

PRE-PART 1

UR5e Model Setup and Cross-Validation

From Manufacturer Data to a Consistent Analytical and Physics-Simulation Model

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Purpose. Establish a defensible common model between a Python rigid-body implementation and NVIDIA Isaac Sim before using either environment for controller design, tuning, and performance comparison.

Abstract

This report documents the model-establishment phase that precedes controller design for a UR5e manipulator. The objective was not merely to import a robot description into a simulator, but to demonstrate that the analytical Python model and the Isaac Sim articulation represented the same nominal rigid-body system under the same coordinate, inertial, gravity, and boundary-condition conventions. The work therefore treated simulation setup itself as a validation problem.

The analytical model was constructed from the nominal UR5e kinematic and physical parameters supplied through Universal Robots and the official ROS 2 description package. Denavit-Hartenberg frames were retained for analytical kinematics, while a separate URDF-aligned physical-link representation was used for rigid-body dynamics. This separation was necessary because equivalent robot descriptions need not use identical intermediate frame conventions.

A first comparison produced an apparent 64.7% relative discrepancy in the joint-space mass matrix. Rather than immediately modifying the analytical dynamics, the discrepancy was treated as a diagnostic signal. The simulator was subsequently found to have an inappropriate articulation/base configuration, an undefined effective gravity setting, and zero joint-drive stiffness and damping. After establishing a fixed-base six-DOF articulation, explicitly setting gravity to $[0, 0, -9.81]$ m/s², tuning the validation drives, and verifying that commanded configurations were statically maintained, the cross-validation was repeated.

At the validation configuration $q = [10, -40, 60, -30, 45, 20]$ deg, the Python and Isaac Sim models showed close agreement in physical-link poses, joint axes, geometric Jacobian, joint-space mass matrix, and generalized gravity forces. The shoulder gravity torque, for example, was -49.685494 Nm in Python and -49.681679 Nm in Isaac Sim. The final result supports the use of the analytical model as a reference for subsequent nonlinear dynamics characterization and controller studies, while also defining the limits of that claim: the work validates a nominal rigid-body UR5e without payload, gripper, structural flexibility, backlash, or robot-specific calibration.

Key Engineering Principle

Model credibility. A disagreement between an analytical model and a simulator does not, by itself, identify which model is wrong. The simulation assumptions and boundary conditions must be validated before a mathematically consistent model is altered to match simulation output.

Validation Scope at a Glance

Quantity / condition	Purpose	Outcome
Manufacturer / ROS parameters	Establish traceable nominal robot definition	Verified and reconciled
Frame conventions	Separate DH kinematics from URDF physical-link dynamics	Reconciled
Fixed-base articulation	Remove unintended floating-base DOFs	6 DOF; 6×6 mass matrix
Units / up axis / gravity	Establish common world physics	1 m/unit, Z-up, $g = -9.81$ m/s ² Z
Joint-drive hold	Ensure measured q equals intended q during validation	Stable at $q=0$ and q_{test}
Physical-link poses	Validate transform propagation	Close agreement
Joint axes / Jacobian	Validate differential kinematics	Close agreement
Mass matrix $M(q)$	Validate inertial dynamics	Close agreement
Gravity vector $g(q)$	Validate mass, CoM, gravity and sign convention	Close agreement

1. Motivation and Validation Philosophy

Controller design is meaningful only when the plant model being controlled is sufficiently well defined. For this project, that requirement is stronger than confirming that a UR5e mesh appears correctly in a simulator. A controller designed against one frame convention, one inertia representation, or one base condition can appear to work while actually compensating for a modeling inconsistency. The Pre-Part 1 task was therefore defined as a model-credibility exercise.

I assumed that the nominal manufacturer and ROS descriptions were the appropriate starting point for a reproducible study, but I did not assume that the analytical and simulated representations would become equivalent automatically. I expected discrepancies to arise primarily from coordinate conventions, importer behavior, physics-scene assumptions, or actuator settings. For that reason, validation was deliberately ordered from low-level geometric quantities to higher-level dynamic quantities.

The validation hierarchy was: (1) source parameters, (2) physical-link transforms, (3) joint origins and axes, (4) geometric Jacobian, (5) joint-space mass matrix, and (6) gravity generalized forces. This ordering matters. A mass-matrix disagreement is difficult to diagnose directly because it can be caused by link masses, CoM locations, inertia tensors, inertia orientations, joint axes, frame transforms, base conditions, or even comparison at different configurations. By validating the lower-level dependencies first, the diagnostic search space is reduced before the higher-level dynamics are judged.

A second principle was to distinguish a simulation artifact from a physical-model property. Joint-drive gains, for example, were needed to hold a configuration under gravity during cross-validation, but those gains are not treated as measured UR5e servo parameters. They are part of the validation environment. The later controller-comparison study must therefore prevent these hidden drives from masking the behavior of the controllers under investigation.

UR5e Analytical-to-Simulation Validation Workflow

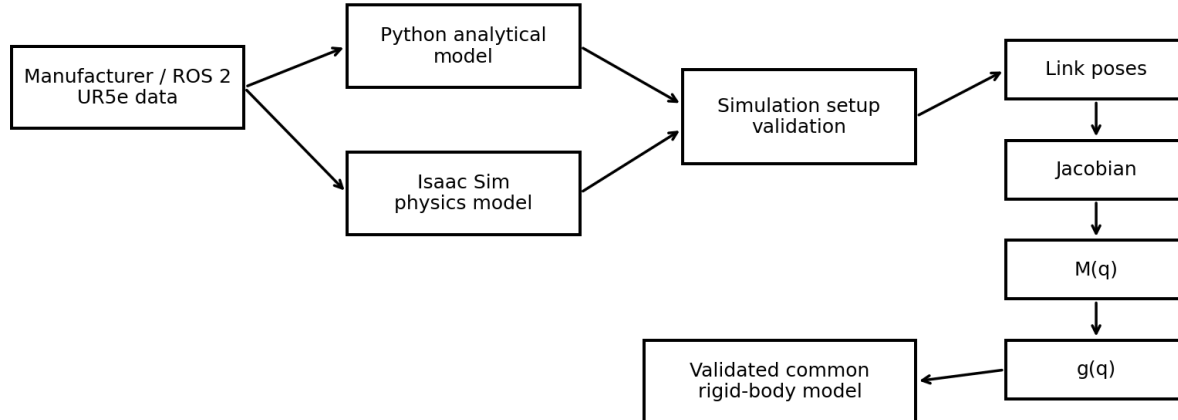


Figure 1. Analytical-to-simulation validation workflow

2. Reference Robot and Source Hierarchy

The reference robot is the Universal Robots UR5e, modeled as a six-revolute-joint serial manipulator. The study uses the nominal manufacturer model rather than calibration parameters extracted from a specific physical robot. This distinction is important: the official ROS 2 description can be parameterized with calibration values from an individual robot, whereas this project is intended to establish a reproducible nominal reference model.

2.1 Nominal kinematic parameters

Joint	a [m]	d [m]	α [rad]
1	0	0.1625	$+\pi/2$
2	-0.4250	0	0
3	-0.3922	0	0
4	0	0.1333	$+\pi/2$
5	0	0.0997	$-\pi/2$
6	0	0.0996	0

The nominal DH parameters above were used for the analytical kinematic chain. The joint coordinates q are state variables and are therefore not embedded as constants in the parameter module.

2.2 Physical parameters used by the rigid-body model

Link	Mass [kg]	URDF-aligned CoM [m]
1 / shoulder	3.761	[0, -0.00193, -0.02561]
2 / upper arm	8.058	[-0.2125, 0, 0.11336]
3 / forearm	2.846	[-0.2422, 0, 0.02650]
4 / wrist 1	1.370	[0, -0.01634, -0.00180]
5 / wrist 2	1.300	[0, 0.01634, -0.00180]
6 / wrist 3	0.365	[0, 0, -0.001159]
Moving-link total	17.700	—

The six moving-link masses sum to 17.7 kg. This is not expected to equal the complete product arm weight reported in the technical specification (20.6 kg including cable), because the rigid-body model and the product-level mass statement have different accounting boundaries. I explicitly avoided distributing the difference across the six links, because doing so would introduce untraceable inertial assumptions.

The CoM coordinates visible on the manufacturer dynamics page differ in sign and component ordering from the ROS physical-parameter representation. I initially expected these values to be numerically identical. I diagnosed the discrepancy as a frame-convention issue rather than a physical disagreement after tracing the URDF transforms and validating the imported link poses. The ROS/URDF-aligned values above were therefore used for rigid-body dynamics.

2.3 Inertia tensors

Link	I_{xx}	I_{yy}	I_{zz}	Selected products of inertia [kg·m ²]
1	0.00700210	0.00648091	0.00657286	$I_{xy}=0.00000073$; $I_{xz}=-0.00001053$; $I_{yz}=0.00049994$

Link	I_{xx}	I_{yy}	I_{zz}	Selected products of inertia [kg·m ²]
2	0.01505885	0.33388086	0.33247207	$I_{xy}=-0.00005400$; $I_{xz}=0.00000563$; $I_{yz}=-0.00000181$
3	0.00399632	0.07879254	0.07848510	$I_{xy}=-0.00001365$; $I_{xz}=0.00137272$; $I_{yz}=-0.00000660$
4	0.00165491	0.00135962	0.00126279	$I_{xy}=-0.00000282$; $I_{xz}=-0.00000438$; $I_{yz}=0.00010157$
5	0.00135617	0.00127827	0.00096614	$I_{xy}=-0.00000274$; $I_{xz}=0.00000444$; $I_{yz}=-0.00005048$
6	0.00018694	0.00018908	0.00025756	$I_{xy}=0.00000006$; $I_{xz}=-0.00000017$; $I_{yz}=-0.00000092$

All six inertia tensors were checked for symmetry and positive definiteness before use. Their eigenvalues were positive. This was a necessary but not sufficient check: a positive-definite inertia tensor can still be expressed in the wrong frame. The later mass-matrix comparison therefore served as a stronger system-level validation.

3. Analytical Model Architecture

3.1 Two representations for two purposes

The analytical implementation intentionally maintains two compatible representations. Standard DH transforms are used for forward/inverse kinematics and geometric reasoning, while a URDF-aligned physical-link transform chain is used for link CoM and inertia propagation. This avoids forcing the URDF into DH conventions or, conversely, using DH frames as if they were physical inertia frames.

$$T_0^6(q) = A_1(q_1) A_2(q_2) \cdots A_6(q_6)$$

The DH implementation was independently checked at $q = 0$ and by finite differences. At $q = 0$, the calculated TCP position was $[-0.8172, -0.2329, 0.0628]$ m, and the rotation matrix satisfied orthonormality and $\det(R)=1$. Finite-difference checks of all joints produced errors on the order of 10^{-7} or smaller.

3.2 URDF-aligned physical-link chain

Inspection of the generated robot description revealed a fixed transformation between `base_link` and `base_link_inertia` equivalent to $R_z(\pi)$. This transformation explains why direct comparison of DH-frame coordinates and physical-link coordinates can appear sign-reversed while still representing the same robot. The physical-link transform implementation therefore begins with this base-frame rotation and then follows the URDF joint origins and rotations.

I expected the $q=0$ comparison to be necessary but not sufficient. A frame implementation can accidentally agree in a special configuration. Consequently, the physical-link transforms were later validated again at a nonzero six-joint configuration.

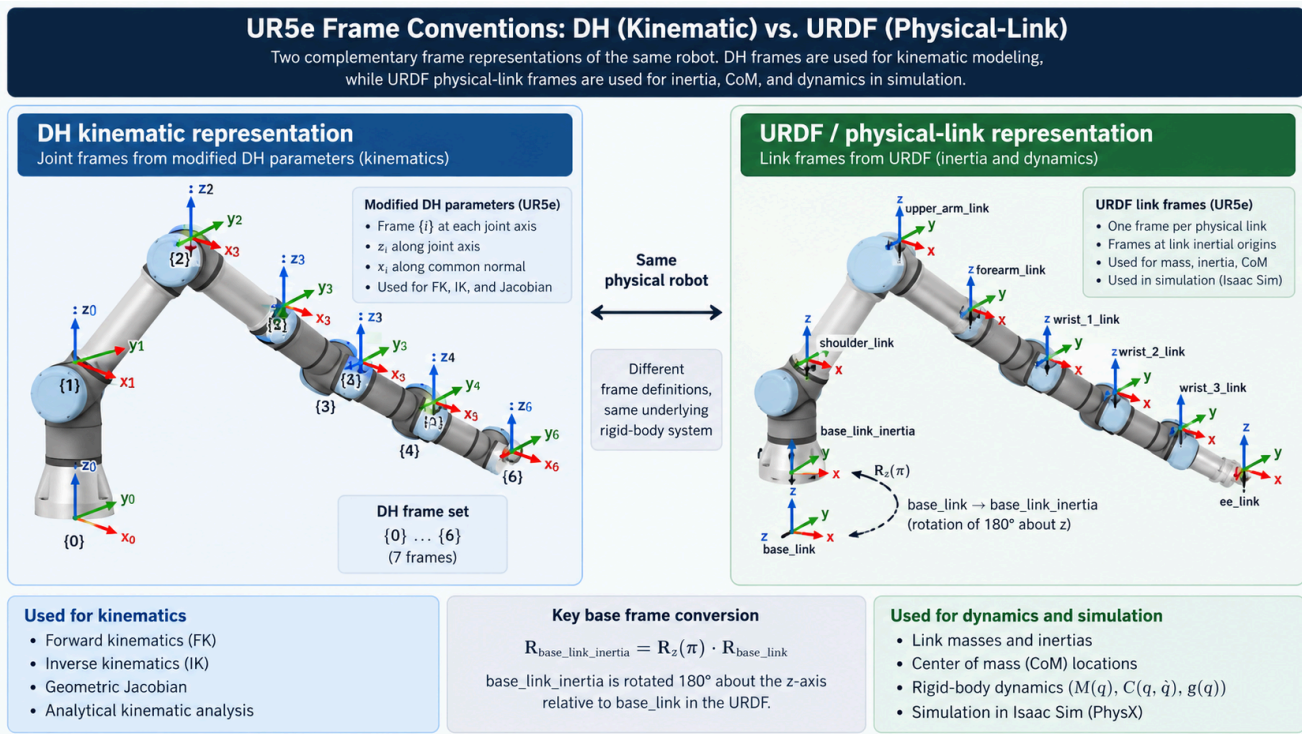


Figure 2. DH frames versus URDF physical-link frames

4. Isaac Sim Model Setup and Diagnostic Corrections

4.1 URDF generation and import

The UR5e description was generated from the official ROS 2 `ur_description` package using the UR5e joint-limit, default-kinematics, physical-parameter, and visual-parameter YAML files. The resulting URDF was imported into Isaac Sim 6.1. The imported articulation contained the expected six revolute joints and preserved the physical mass, CoM, and inertia information.

Direct inspection of the imported USD mass properties confirmed the six moving-link masses and CoM values. The simulator stored principal moments and principal-axis orientations rather than simply reproducing the full local inertia matrix entry-for-entry; this is physically equivalent when the principal-axis transform is applied correctly.

4.2 Fixed-base diagnosis

The first major diagnostic signal was the size of the simulator mass matrix. A fixed six-DOF industrial manipulator should produce a 6×6 joint-space mass matrix. The imported model initially returned a 12×12 matrix, indicating that six floating-base coordinates were being included.

I diagnosed the 12×12 result as evidence that the simulator was not yet representing the intended boundary condition. A temporary world-to-base fixed joint physically prevented base motion, but this solution was incomplete: the Gain Tuner subsequently reported a robot-schema relationship error because the manually created joint did not conform to the articulation structure expected by the toolchain.

I considered 'the robot no longer moves at the base' insufficient as a validation criterion. The articulation also needed to be internally consistent with Isaac Sim's robot schema and physics representation. The temporary joint was therefore removed, and the fixed-base articulation was repaired using Isaac Sim's fixed-base articulation utility. The resulting articulation returned the expected 6×6 mass matrix and the Gain Tuner opened without the previous schema error.

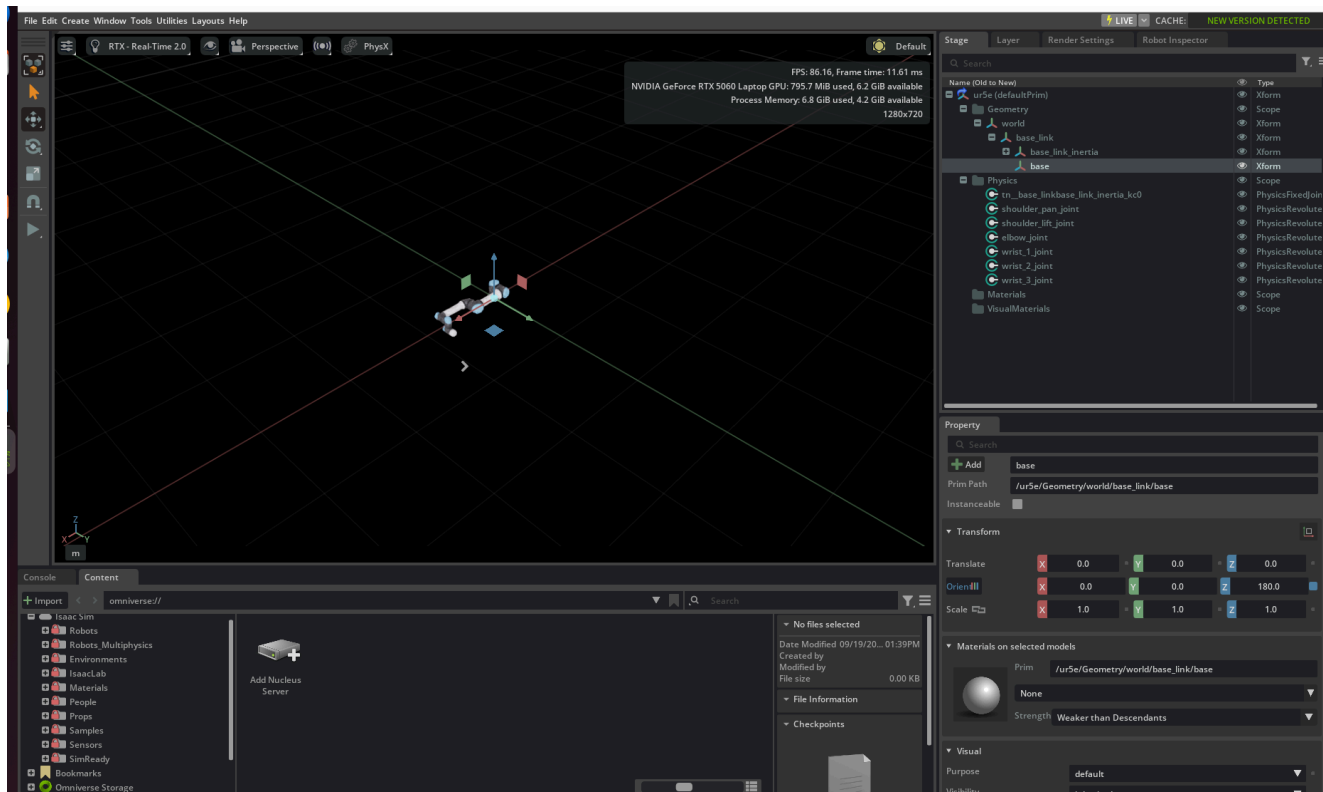


Figure 3. Final fixed-base articulation structure

4.3 World physics and gravity

Setting	Observed before correction	Final value
Stage up axis	Z	Z
Meters per unit	1.0	1.0
Gravity direction	(0, 0, 0)	(0, 0, -1)
Gravity magnitude	-inf	9.81 m/s ²

The world coordinate convention and metric scale were already appropriate, but gravity was not represented by a normal finite vector. I therefore explicitly authored a downward Z gravity field of 9.81 m/s². The analytical model uses the identical vector $g = [0, 0, -9.81]$ m/s². This step was essential because gravity-vector validation is meaningless unless both models share the same world direction and magnitude.

4.4 Joint drives and static configuration control

The imported revolute joints had finite drive-force limits but zero stiffness and zero damping. In that condition, a target position is not actively maintained. Under gravity, the robot moved away from the intended validation configuration. This meant that even a mathematically correct comparison could inadvertently evaluate Python at q_{target} while Isaac Sim was physically at $q_{\text{actual}} \neq q_{\text{target}}$.

The joint effort limits were retained as imported: 150 for the first three joints and 28 for the three wrist joints. These values are treated here as URDF/simulation drive limits, not as independently verified motor torque ratings.

Joint	Kp	Kd	Max force
shoulder_pan_joint	445.41058	0.47260	150
shoulder_lift_joint	7106.18213	7.53989	150
elbow_joint	39952.72266	42.39116	150
wrist_1_joint	1209.48828	1.28331	28
wrist_2_joint	341.18130	0.36200	28
wrist_3_joint	11.75560	0.01247	28

For validation, the Isaac Sim Gain Tuner was used in natural-frequency mode with 300 Hz and damping ratio $\zeta = 1.0$ as the starting condition. This follows NVIDIA's manipulator setup guidance, which uses the same starting values for a UR manipulator. The Gain Tuner converts the requested natural frequency and damping ratio into joint-specific stiffness and damping using the effective joint inertia.

$$\omega_n = \sqrt{(K_p / m_{\text{eff}})}, \quad \zeta = K_d / (2 m_{\text{eff}} \omega_n)$$

These gains are not claimed to represent the bandwidth of the physical UR5e controller. Their purpose in this phase is narrower: hold a commanded configuration reproducibly under gravity while the rigid-body models are compared. In Part 1B, the controller study must ensure that these validation drives do not mask the dynamics of PID, LQR, MPC, or other controllers being evaluated.

A ± 10 deg step test on the shoulder-lift joint produced a peak position error of 0.036 deg and RMS error of 0.015 deg. More importantly, the final static-hold tests showed that the robot remained stationary at both $q=0$ and the nonzero validation configuration.

Condition	Target [deg]	Measured / observed
Home configuration	[0, 0, 0, 0, 0, 0]	J2 = -0.01259 deg; all other errors < 0.001 deg
Validation configuration	[10, -40, 60, -30, 45, 20]	[9.999999, -39.999996, 60, -30, 45, 19.999998] deg

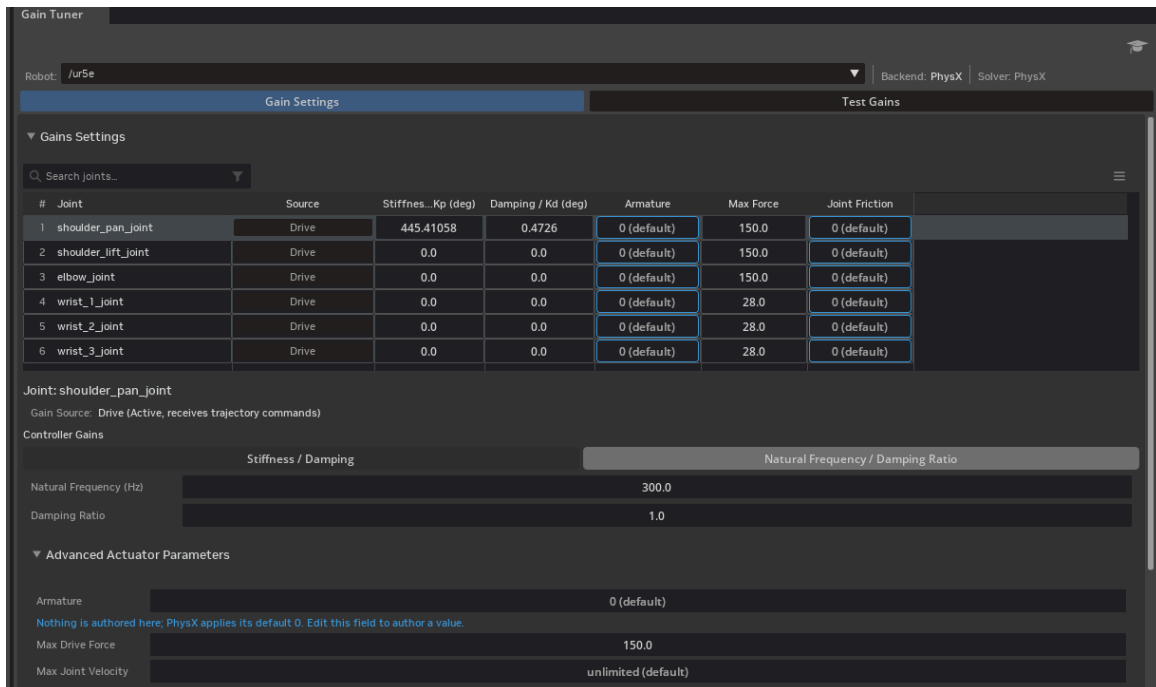


Figure 4. Isaac Sim Gain Tuner configuration

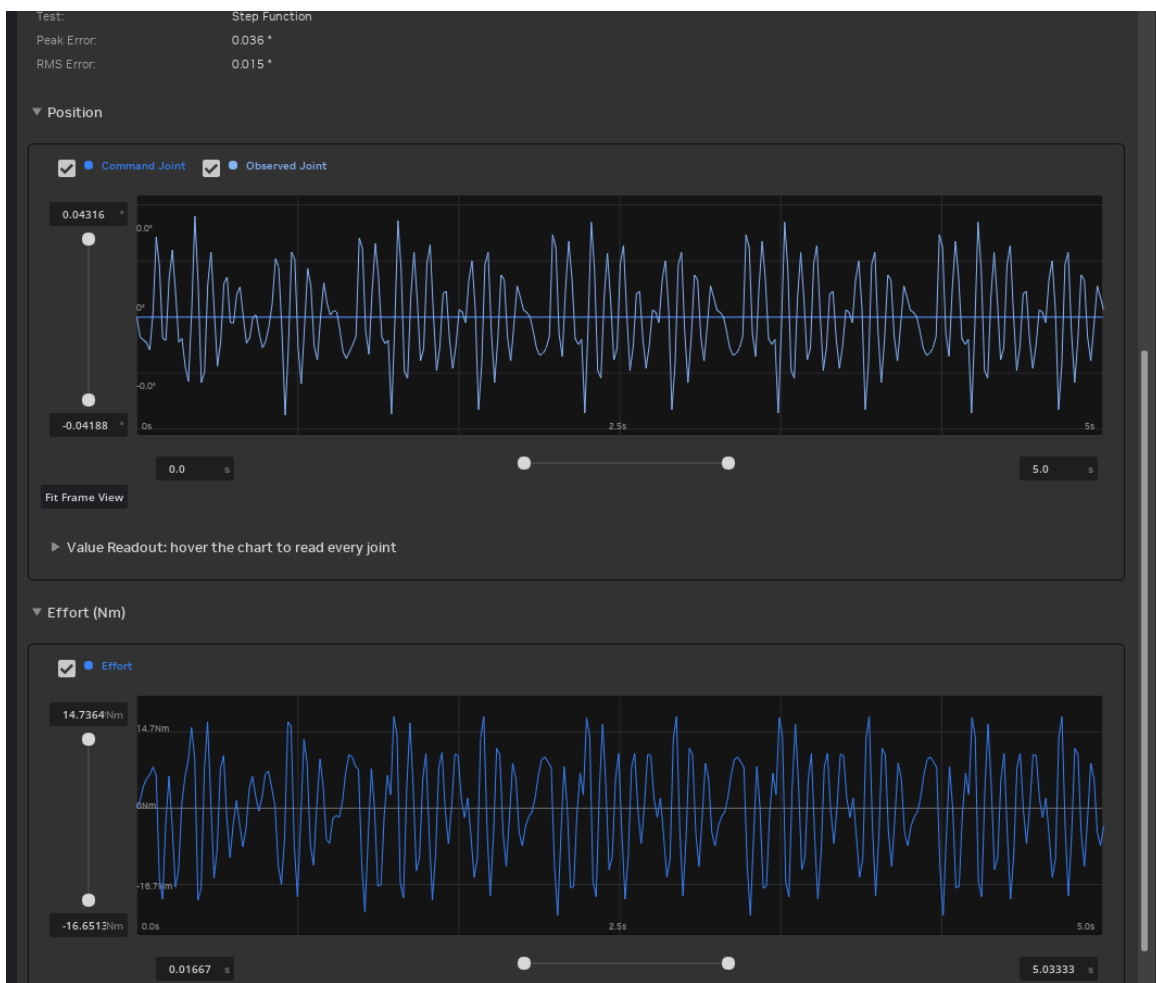


Figure 5. Shoulder-lift drive validation

5. Cross-Validation Methodology

Cross-validation used the common nonzero configuration $q_{\text{test}} = [10, -40, 60, -30, 45, 20]$ deg. A nonzero configuration was selected to exercise all joint rotations and reduce the chance of accepting a transform implementation that only agrees at a special home pose.

Comparison rule. For every quantity, the simulator configuration was first verified to be stationary at q_{test} . The analytical result was then evaluated at the same q . Differences were interpreted only after frame, ordering, units, and sign conventions had been checked.

5.1 Physical-link pose validation

Link	Python position [m]	Isaac position [m]
shoulder_link	[0, 0, 0.162500]	[0, 0, 0.162500]
upper_arm_link	[0, 0, 0.162500]	[0, 0, 0.162500]
forearm_link	[0.320623, 0.056534, 0.435685]	[0.320565, 0.056517, 0.435756]
wrist_1_link	[0.660424, 0.251807, 0.301544]	[0.660402, 0.251788, 0.301705]
wrist_2_link	[0.677474, 0.254813, 0.203359]	[0.677478, 0.254799, 0.203525]
wrist_3_link	[0.733548, 0.336215, 0.215589]	[0.733552, 0.336198, 0.215773]

The wrist_3 frame-origin position differs by approximately 0.185 mm in Euclidean norm, while the intermediate links show similarly small discrepancies. Rotation matrices also agreed closely; representative element differences were on the order of 10^{-4} to 10^{-3} . The nonzero-pose agreement established that the URDF-aligned transform propagation was not merely matching at $q=0$.

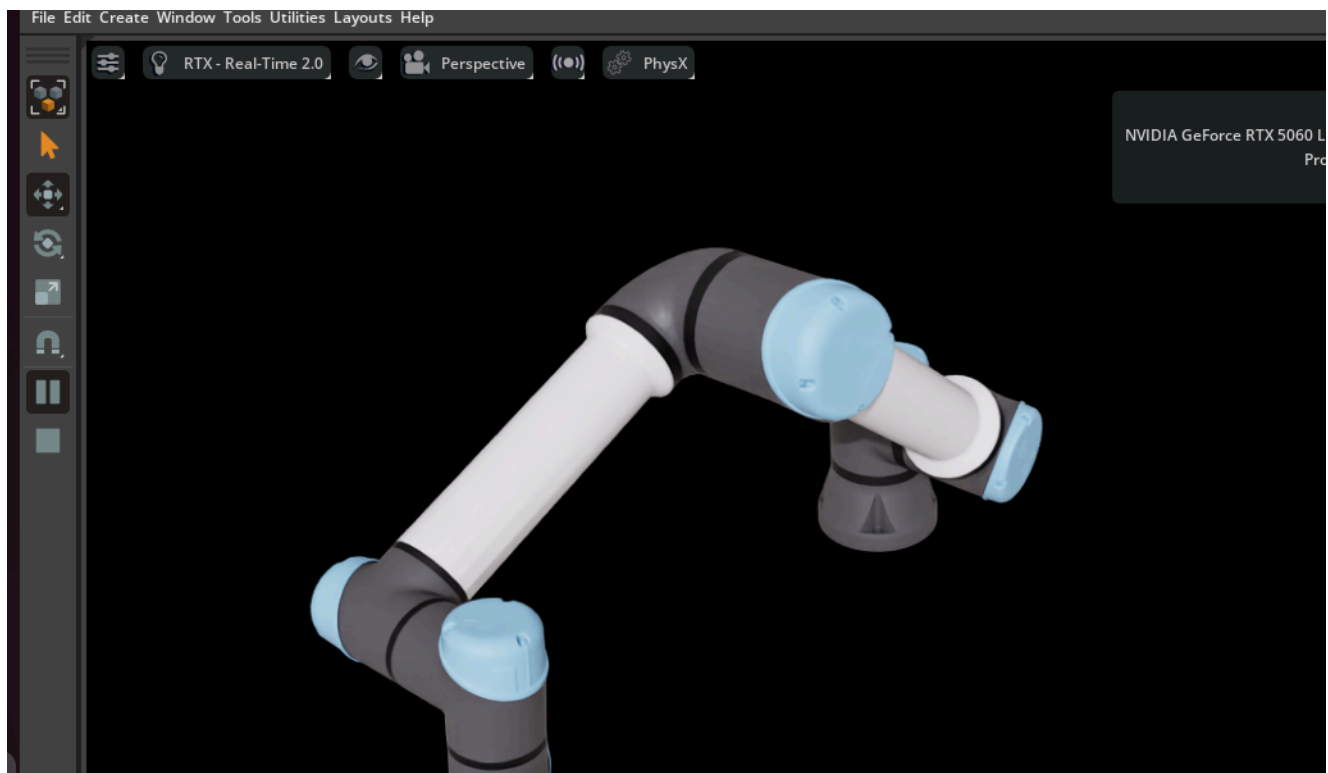


Figure 6. Common validation pose in Python and Isaac Sim

5.2 Joint-axis and Jacobian validation

For a revolute joint i , the geometric Jacobian was assembled from the world-frame joint axis z_i , joint origin p_i , and evaluation point p_e :

$$J_{v,i} = z_i \times (p_e - p_i), \quad J_{\omega,i} = z_i$$

The wrist_3-link Jacobian from the analytical physical-link model was compared with Isaac Sim's articulation Jacobian. Representative angular-axis columns are shown below.

Joint	Python angular axis	Isaac angular axis
J1	[0, 0, 1]	[0, 0, 1]
J2	[-0.173648, 0.984808, 0]	[-0.173630, 0.984811, 0]
J5	[0.171010, 0.030154, -0.984808]	[0.171270, 0.030197, -0.984761]
J6	[0.562997, 0.817287, 0.122788]	[0.562986, 0.817266, 0.122975]

The linear Jacobian showed the same level of agreement. For example, the J1 linear column was [-0.336215, 0.733548, 0] in Python and [-0.335253, 0.732898, 0] in Isaac Sim. This comparison is important because it validates differential kinematics and joint-axis propagation independently of the inertia calculations.

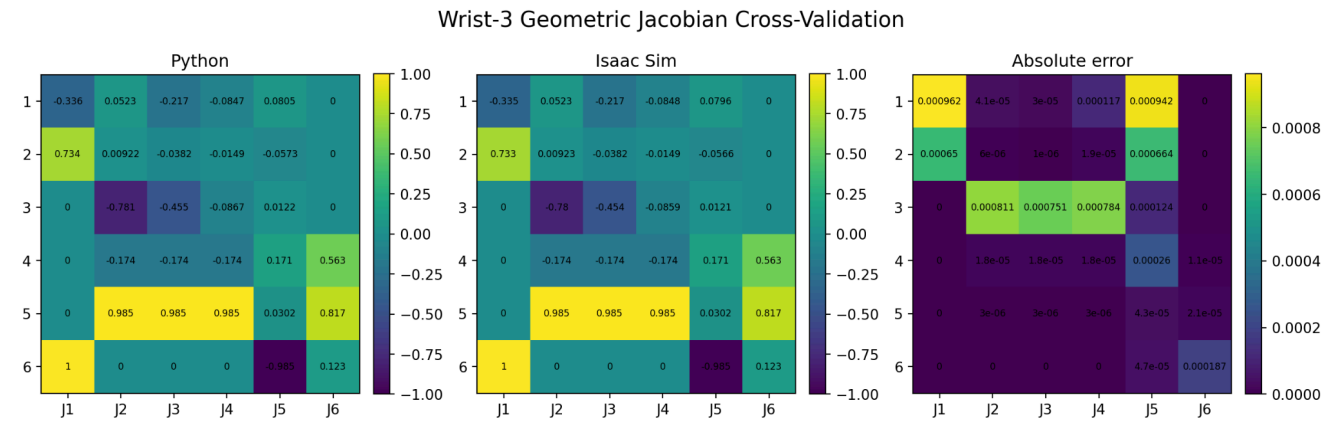


Figure 7. Jacobian cross-validation

5.3 Joint-space mass matrix validation

The analytical mass matrix was assembled by summing translational and rotational kinetic-energy contributions of all six moving links:

$$M(q) = \sum_i [m_i J_{v,i}^T J_{v,i} + J_{\omega,i}^T (R_i I_i R_i^T) J_{\omega,i}]$$

Before the simulator was corrected, the mass-matrix comparison showed an apparent relative discrepancy of approximately 64.7%. This was initially compatible with several hypotheses: an incorrect inertia-frame transformation, a joint-axis convention error, a floating-base mismatch, or comparison at a moving configuration. I initially suspected a frame or inertia-tensor transformation issue because those are common failure modes in rigid-body implementations. However, I considered changing the analytical model premature because its FK and Jacobian had already passed independent checks.

After the simulator boundary conditions and static-hold behavior were corrected, the mass matrices agreed closely:

Element	Python	Isaac	Absolute difference
M11	3.103413	3.103188	0.000225

Element	Python	Isaac	Absolute difference
M22	3.342034	3.342024	0.000010
M23	1.208672	1.208643	0.000029
M33	0.833906	0.833856	0.000050
M44	0.021196	0.021143	0.000053
M66	0.000258	0.000258	≈0

Both matrices were exactly symmetric to the reported numerical precision. Their eigenvalue spectra were also closely aligned:

Eigenvalue	Python	Isaac
λ_1	0.000255	0.000255
λ_2	0.003714	0.004028
λ_3	0.013381	0.013334
λ_4	0.344965	0.344952
λ_5	3.091497	3.091090
λ_6	3.852034	3.852161

The second eigenvalue shows the largest relative difference because its magnitude is small; the absolute difference remains approximately 3.14×10^{-4} . The dominant inertial modes and the individual matrix elements agree closely. The result strongly supports consistency of the link masses, CoM positions, inertia orientations, joint axes, and physical-link transforms.

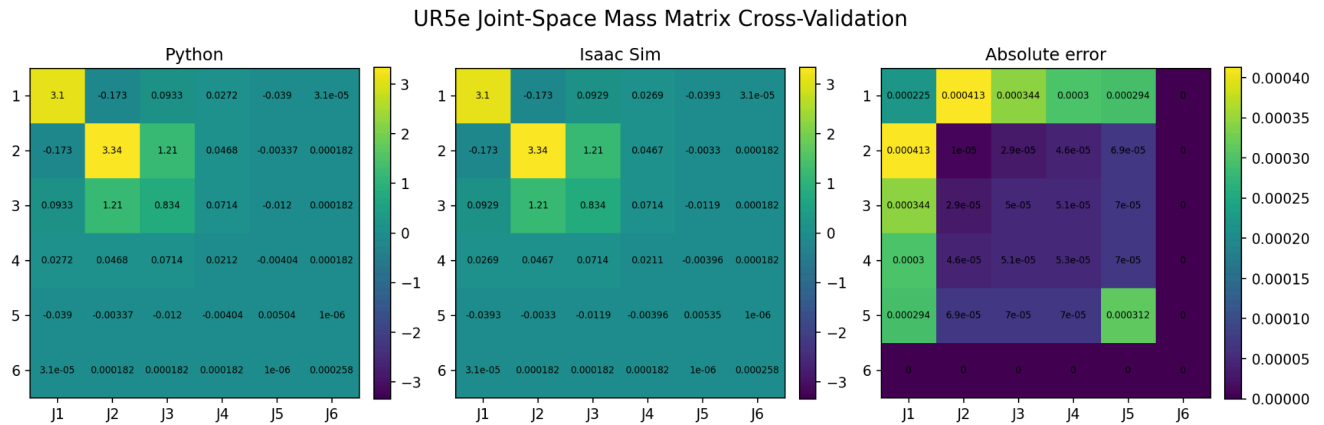


Figure 8. Mass-matrix cross-validation after simulation correction

5.4 Gravity generalized-force validation

The analytical gravity vector was computed independently from potential energy rather than copied from simulator output. For gravity vector $\mathbf{g}_{\text{world}} = [0, 0, -9.81] \text{ m/s}^2$ and link CoM position $\mathbf{p}_{c,i}(\mathbf{q})$, the potential energy was defined as:

$$U(\mathbf{q}) = -\sum_i m_i \mathbf{g}_{\text{world}}^T \mathbf{p}_{c,i}(\mathbf{q}), \quad \mathbf{g}(\mathbf{q}) = \partial U / \partial \mathbf{q}$$

A centered numerical derivative of potential energy was used as the reference implementation. The sign convention corresponds to the manipulator equation $\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$.

Joint	Python [Nm]	Isaac [Nm]	Absolute difference [Nm]
J1	0.000000	0.000008	0.000008
J2	-49.685494	-49.681679	0.003815
J3	-18.034631	-18.036629	0.001998
J4	-0.707495	-0.707961	0.000466
J5	+0.068868	+0.068971	0.000103
J6	0.000000	0.000000	0.000000

The largest absolute difference occurred at the shoulder-lift joint: 0.003815 Nm, approximately 0.0077% of the Python magnitude. The signs also agreed directly, confirming that the analytical convention and the simulator's reported generalized gravity forces were compatible. J1 is approximately zero because its rotation axis is aligned with gravity, while the shoulder and elbow carry the dominant gravitational loading.

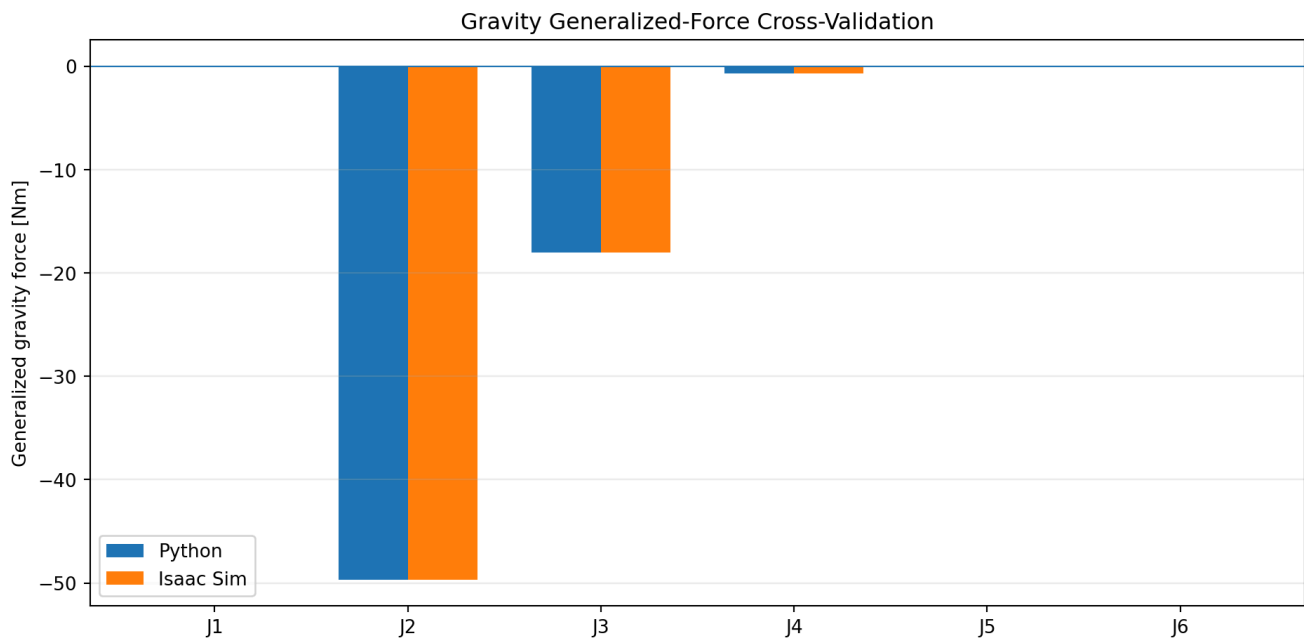


Figure 9. Gravity generalized-force comparison

6. Diagnostic Case Study: Why the First Dynamics Comparison Failed

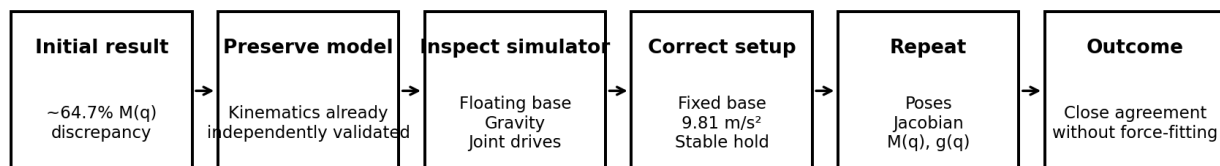
The most instructive result of this phase was not the final agreement; it was the failure of the first comparison. The initial 64.7% mass-matrix discrepancy could easily have triggered a rewrite of the analytical dynamics. Instead, it was treated as evidence that at least one assumption shared by the comparison was false.

I diagnosed the problem in layers. First, the 12×12 simulator mass matrix contradicted the intended six-DOF fixed manipulator and exposed the floating-base interpretation. Second, a manual fixed joint solved physical base motion but introduced a schema inconsistency, demonstrating that a visually stationary robot is not necessarily a correctly formed articulation. Third, the PhysicsScene showed no normal finite gravity definition. Fourth, the imported joint drives had $K_p = K_d = 0$, so the robot could not hold the commanded pose under gravity. This last point was especially important because dynamics were being compared at a nominal q even though the simulator could be evolving away from that q .

Several possible fixes existed at each stage. I could have forced the analytical model to reproduce the initial simulator output, increased arbitrary joint gains, or ignored the articulation-schema warning once the base appeared stationary. I deliberately did not choose those shortcuts. I considered consistency of the physical assumptions more important than obtaining numerical agreement quickly. Only after the simulator represented a fixed manipulator, shared the same gravity convention, and demonstrably held the requested configuration did I repeat the model comparison.

The subsequent agreement of pose, Jacobian, $M(q)$, and $g(q)$ changed the interpretation of the original discrepancy. It was not evidence that the analytical rigid-body model was fundamentally incorrect. It was evidence that the simulation state used as a reference had not yet been validated.

Diagnostic Decision Path: From Disagreement to Validated Agreement



Validate simulation assumptions before changing a consistent analytical model.

Figure 10. Diagnostic decision path from discrepancy to validated model

7. Assumptions, Boundaries, and What Has Not Been Validated

The present validation establishes consistency between two nominal rigid-body representations. It does not establish that either model reproduces every behavior of a particular physical UR5e. The distinction is important for later controller conclusions.

Boundary / assumption	Implication
Nominal UR5e geometry	Robot-specific factory calibration parameters are not included.
No payload or gripper	Validated $M(q)$ and $g(q)$ apply to the bare modeled manipulator.
Rigid links and ideal joints	Structural flexibility, backlash, gearbox compliance, friction nonlinearities, and actuator dynamics are outside the present model.
Validation drives are simulation infrastructure	The tuned Isaac gains are not identified physical UR5e servo gains and must not be allowed to dominate later controller comparisons.
One primary nonzero validation pose	Agreement at q_{test} is strong evidence of consistency but is not a proof over the entire workspace.
URDF/simulator effort limits retained	150/28 values are treated as model limits; they are not independently asserted here as measured motor torque capabilities.
No contact dynamics	Ground, table, grasp, collision, and object interaction are outside Pre-Part 1.

A logical extension, if stronger global evidence is required, is to automate the same comparison over a set of configurations spanning the workspace and report distributions of pose, Jacobian, mass-matrix, and gravity-vector errors. That extension is not required to proceed with Part 1A, but it would strengthen a publication-oriented validation study.

8. Conclusions and Interface to Part 1

Pre-Part 1 established a common nominal UR5e model across the analytical Python implementation and Isaac Sim. The work resolved the distinction between DH and URDF physical-link frames, verified physical parameters, repaired the fixed-base articulation, established the world gravity convention, configured validation-only joint drives, and demonstrated static pose retention before comparing dynamic quantities.

The final cross-validation showed close agreement in physical-link poses, joint axes, the geometric Jacobian, the joint-space mass matrix, and the generalized gravity vector. The agreement is significant because these quantities test different layers of the model. Pose agreement validates transform propagation; Jacobian agreement validates joint-axis and differential-kinematic consistency; mass-matrix agreement tests masses, CoM locations, inertia tensors, inertia orientations, and kinematic propagation together; and gravity-vector agreement independently tests masses, CoM positions, gravity direction, and sign convention.

The main engineering conclusion is methodological: simulation validation must precede the use of simulation as a reference. The initial large discrepancy was not resolved by forcing the analytical model toward the simulator. It was resolved by identifying which assumptions made the comparison invalid. This establishes a more defensible foundation for the next phase of the project.

Part 1A can now proceed from a validated rigid-body baseline to nonlinear joint-space dynamics characterization, including completion and verification of $C(q, \dot{q})$, configuration-dependent dynamic behavior, linearization, pole/eigenvalue analysis, and transient response. Part 1B can then compare PID, computed-torque control, LQR, and MPC while explicitly separating the controllers under study from the simulation drives used only for model-validation setup.

9. Recommended Figures for the Final Portfolio Version

Fig.	Graphic	Why it matters
1	Validation workflow	Shows the logic of the entire study immediately.
2	DH vs URDF frames	Explains the most important frame-convention issue.
3	Final Isaac articulation hierarchy	Documents fixed-base model construction.
4	Gain Tuner settings	Documents the validation-drive environment.
5	Step response	Provides measured evidence that the pose can be held/tracked.
6	Python vs Isaac q_test pose	Visual confirmation of common configuration.
7	Jacobian heatmaps	Compact differential-kinematics validation.
8	Mass-matrix heatmaps/error	Core dynamics-validation figure; highest priority.
9	Gravity torque comparison	Clear quantitative dynamics comparison.
10	Diagnostic decision flow	Shows engineering reasoning rather than only final success.

References

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Appendix A. Numerical Reference Data

A.1 Python mass matrix at q_{test}

	J1	J2	J3	J4	J5	J6
J1	3.103413	-0.173015	0.093256	0.027184	-0.039033	0.000031
J2	-0.173015	3.342034	1.208672	0.046755	-0.003372	0.000182
J3	0.093256	1.208672	0.833906	0.071411	-0.011963	0.000182
J4	0.027184	0.046755	0.071411	0.021196	-0.004035	0.000182
J5	-0.039033	-0.003372	-0.011963	-0.004035	0.005039	0.000001
J6	0.000031	0.000182	0.000182	0.000182	0.000001	0.000258

A.2 Isaac Sim mass matrix at q_{test}

	J1	J2	J3	J4	J5	J6
J1	3.103188	-0.173428	0.092912	0.026884	-0.039327	0.000031
J2	-0.173428	3.342024	1.208643	0.046709	-0.003303	0.000182
J3	0.092912	1.208643	0.833856	0.071360	-0.011893	0.000182
J4	0.026884	0.046709	0.071360	0.021143	-0.003965	0.000182
J5	-0.039327	-0.003303	-0.011893	-0.003965	0.005351	0.000001
J6	0.000031	0.000182	0.000182	0.000182	0.000001	0.000258

A.3 Representative wrist_3 Jacobians

Python	J1	J2	J3	J4	J5	J6
r1	-0.336215	0.052282	-0.216752	-0.084650	0.080534	0.000000
r2	0.733548	0.009219	-0.038219	-0.014926	-0.057314	0.000000
r3	0.000000	-0.780787	-0.455218	-0.086671	0.012223	0.000000
r4	0.000000	-0.173648	-0.173648	-0.173648	0.171010	0.562997
r5	0.000000	0.984808	0.984808	0.984808	0.030154	0.817287
r6	1.000000	0.000000	0.000000	0.000000	-0.984808	0.122788

Isaac	J1	J2	J3	J4	J5	J6
r1	-0.335253	0.052323	-0.216782	-0.084767	0.079592	0.000000
r2	0.732898	0.009225	-0.038220	-0.014945	-0.056650	0.000000
r3	0.000000	-0.779976	-0.454467	-0.085887	0.012106	0.000000
r4	0.000000	-0.173630	-0.173630	-0.173630	0.171270	0.562986
r5	0.000000	0.984811	0.984811	0.984811	0.030197	0.817266
r6	1.000000	0.000000	0.000000	0.000000	-0.984761	0.122975